Given a predefined block sparsity mask $\mathbf{M} \in \{0,1\}^{N/B_r \times N/B_c}$ we can easily adapt Algorithm 1 to only compute the nonzero blocks of the attention matrix. The algorithm is identical to Algorithm 1, except we skip zero blocks. We reproduce the algorithm description in Algorithm 5 in Appendix B.

We also analyze the IO complexity of block-sparse FlashAttention.

Proposition 4. *Let N be the sequence length, d be the head dimension, and M be size of SRAM with $d \leq M \leq Nd$. Block-sparse* FlashAttention *(Algorithm 5) requires $\Theta(Nd + N^2d^2M^{-1}s)$ HBM accesses where s is the fraction of nonzero blocks in the block-sparsity mask.*

We see that applying block-sparsity yields a direct improvement by the sparsity to the larger term in the IO complexity. For large sequence lengths N, s is often set to $N^{-1/2}$ [11] or $N^{-1}\log N$ [3, 17, 92], resulting in $\Theta(N\sqrt{N})$ or $\Theta(N\log N)$ IO complexity. For downstream experiments, we use the fixed butterfly sparsity pattern [17], which has been shown to be able to approximate arbitrary sparsity [16].

In Fig. 2 (right), we validate that as the sparsity increases, the runtime of block-sparse FlashAttention improves proportionally. On the LRA benchmark, block-sparse FlashAttention achieves 2.8× speedup, while performing on par with standard attention (Section 4).

4 Experiments

We evaluate the impact of using FlashAttention to train Transformer models. We validate two claims about training time and model accuracy, and report attention runtime and memory benchmarks.

- **Training Speed.** FlashAttention outperforms the MLPerf 1.1 [58] speed record for BERT by 15%, and speeds up GPT-2 up to 3× over HuggingFace [87] and 1.8× over Megatron [77] over standard Transformers. FlashAttention speeds up the long-range arena (LRA) benchmark 2.4×.
- **Quality.** FlashAttention scales Transformers to longer sequences, yielding higher quality. FlashAttention trains GPT-2 with context length 4K faster than Megatron trains GPT-2 with context length 1K, while achieving 0.7 better perplexity. Modeling longer sequences yields 6.4 points of lift on two long-document classification tasks. Finally, FlashAttention yields the **first Transformer** that can achieve better-than-random performance on the challenging Path-X task (sequence length 16K), and block-sparse FlashAttention yields the **first sequence model** that we know of that can achieve better-than-random performance on Path-256 (sequence length 64K).
- **Benchmarking Attention.** We measure the runtime and memory performance of FlashAttention and block-sparse FlashAttention based on sequence length. We confirm that the memory footprint of FlashAttention scales linearly with seq. length and is up to 3× faster than standard attention for common seq. lengths (up to 2K). We confirm that runtime of block-sparse FlashAttention scales linearly in seq. length and is faster than all existing approximate attention baselines.

Additional experiment details are in Appendix E.

4.1 Faster Models with FlashAttention

BERT. FlashAttention yields the fastest single-node BERT training speed that we know of. We train a BERT-large [22] model with FlashAttention on Wikipedia. Table 1 compares our training time to the implementation from Nvidia that set the training speed record for MLPerf 1.1 [58]. Our implementation is 15% faster.

Table 1: Training time of BERT-large, starting from the same initialization provided by the MLPerf benchmark, to reach the target accuracy of 72.0% on masked language modeling. Averaged over 10 runs on 8×A100 GPUs.

BERT Implementation	Training time (minutes)
Nvidia MLPerf 1.1 [58]	20.0 ± 1.5
FlashAttention (ours)	**17.4** ± 1.4

GPT-2. FlashAttention yields faster training times for GPT-2 [67] on the large OpenWebtext dataset [32] than the widely used HuggingFace [87] and Megatron-LM [77] implementations. Table 2 shows up to 3× end-to-end speedup compared to Huggingface and 1.7× speedup compared to Megatron-LM. FlashAttention

achieves the same perplexity as the other two implementations, as we do not change the model definition. Appendix E includes plots of the validation perplexity throughout training, confirming that FlashAttention is as numerically stable as the baselines and produces the same training / validation curves.

Table 2: GPT-2 small and medium using FlashAttention achieve up to 3× speed up compared to Huggingface implementation and up to 1.7× compared to Megatron-LM. Training time reported on 8×A100s GPUs.

Model implementations	OpenWebText (ppl)	Training time (speedup)
GPT-2 small - Huggingface [87]	18.2	9.5 days (1.0×)
GPT-2 small - Megatron-LM [77]	18.2	4.7 days (2.0×)
GPT-2 small - FlashAttention	18.2	**2.7 days (3.5×)**
GPT-2 medium - Huggingface [87]	14.2	21.0 days (1.0×)
GPT-2 medium - Megatron-LM [77]	14.3	11.5 days (1.8×)
GPT-2 medium - FlashAttention	14.3	**6.9 days (3.0×)**

Long-range Arena. We compare vanilla Transformer (with either standard implementation or FlashAttention) on the long-range arena (LRA [80]) benchmark. We measure accuracy, throughput, and training time of all models. Each task has a different sequence length varying between 1024 and 4096. We follow the implementation and experimental setting in Tay et al. [80]and Xiong et al. [90].[3] Table 3 shows that FlashAttention achieves up 2.4× speed-up compared to standard attention. Block-sparse FlashAttention is faster than all of the approximate attention methods that we have tested.

Table 3: The performance of standard attention, FlashAttention, block-sparse FlashAttention, and approximate attention baselines on the Long-Range-Arena benchmarks.

Models	ListOps	Text	Retrieval	Image	Pathfinder	Avg	Speedup
Transformer	36.0	63.6	81.6	42.3	72.7	59.3	-
FlashAttention	37.6	63.9	81.4	43.5	72.7	59.8	2.4×
Block-sparse FlashAttention	37.0	63.0	81.3	43.6	73.3	59.6	**2.8**×
Linformer [84]	35.6	55.9	77.7	37.8	67.6	54.9	2.5×
Linear Attention [50]	38.8	63.2	80.7	42.6	72.5	59.6	2.3×
Performer [12]	36.8	63.6	82.2	42.1	69.9	58.9	1.8×
Local Attention [80]	36.1	60.2	76.7	40.6	66.6	56.0	1.7×
Reformer [51]	36.5	63.8	78.5	39.6	69.4	57.6	1.3×
Smyrf [19]	36.1	64.1	79.0	39.6	70.5	57.9	1.7×

4.2 Better Models with Longer Sequences

Language Modeling with Long Context. The runtime and memory-efficiency of FlashAttention allow us to increase the context length of GPT-2 by 4× while still running faster than the optimized implementation from Megatron-LM. Table 4 shows that that GPT-2 with FlashAttention and context length 4K is still 30% faster than GPT-2 from Megatron with context length 1K, while achieving 0.7 better perplexity.

Table 4: GPT-2 small with FlashAttention, with 4× larger context length compared to Megatron-LM, is still 30% faster while achieving 0.7 better perplexity. Training time on 8×A100 GPUs is reported.

Model implementations	Context length	OpenWebText (ppl)	Training time (speedup)
GPT-2 small - Megatron-LM	1k	18.2	4.7 days (1.0×)
GPT-2 small - FlashAttention	1k	18.2	**2.7 days (1.7×)**
GPT-2 small - FlashAttention	2k	17.6	3.0 days (1.6×)
GPT-2 small - FlashAttention	4k	**17.5**	3.6 days (1.3×)

Long Document Classification. Training Transformers with longer sequences with FlashAttention improves performance on the MIMIC-III [47] and ECtHR [6, 7] datasets. MIMIC-III contains intensive care unit patient discharge summaries, each annotated with multiple labels. ECtHR contains legal cases from the

[3] LRA accuracy results are known to be highly dependent on the tuning procedure [90]. Our reproduced baselines perform better than as reported in the original comparison [80].

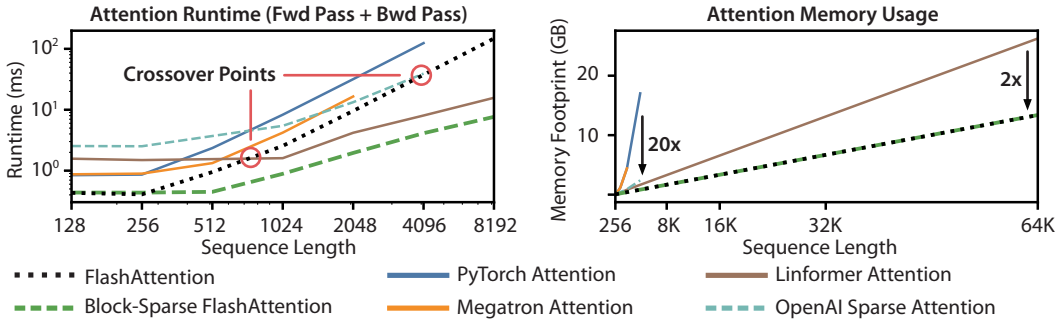

Figure 3: **Left:** runtime of forward pass + backward pass. **Right:** attention memory usage.

European Court of Human Rights, each of which is mapped to articles of the Convention of Human Rights that were allegedly violaged. Both of these datasets contain very long text documents; the average number of tokens in MIMIC is 2,395 tokens, and the longest document contains 14,562 tokens, while the average and longest numbers in ECtHR are 2,197 and 49,392, respectively. We evaluate lift from increasing the sequence length of a pretrained RoBERTa model [56] (we repeat the positional embeddings, as in Beltagy et al. [3]).

Table 5 shows that sequence length 16K outperforms length 512 by 4.3 points on MIMIC, and that length 8K outperforms length 512 by 8.5 points on ECtHR. The discrepancies may be due to subtle distribution shifts: MIMIC-III contains specialized medical text and thus may be more susceptible to a distribution shift in the document length, whereas ECtHR contains general language.

Table 5: Long Document performance (micro F_1) at different sequence lengths using FlashAttention.

	512	1024	2048	4096	8192	16384
MIMIC-III [47]	52.8	50.7	51.7	54.6	56.4	**57.1**
ECtHR [6]	72.2	74.3	77.1	78.6	**80.7**	79.2

Table 6: We report the first Transformer model that can achieve non-random performance on Path-X and Path-256.

Model	Path-X	Path-256
Transformer	✗	✗
Linformer [84]	✗	✗
Linear Attention [50]	✗	✗
Performer [12]	✗	✗
Local Attention [80]	✗	✗
Reformer [51]	✗	✗
SMYRF [19]	✗	✗
FlashAttention	**61.4**	✗
Block-sparse FlashAttention	56.0	**63.1**

Path-X and Path-256. The Path-X and Path-256 benchmarks are challenging tasks from the long-range arena benchmark designed to test long context. The task is to classify whether two points in a black and white 128×128 (or 256×256) image have a path connecting them, and the images are fed to the transformer one pixel at a time. In prior work, all transformer models have either run out of memory, or only achieved random performance [80]. There has been a search for alternative architectures that can model such long context [37]. We present here the first result of Transformer models being able to solve Path-X and Path-256 (Table 6). We pretrain a transformer on Path-64, and then transfer to Path-X by spatially interpolating the positional embeddings. FlashAttention achieves 61.4 accuracy on Path-X. Additionally, block-sparse FlashAttention enables the Transformers to scale to sequence length 64K, achieving 63.1 accuracy[4] on Path-256.

4.3 Benchmarking Attention

We vary sequence length and measure runtime and memory usage of FlashAttention and block-sparse FlashAttention against various attention baselines on one A100 GPU with 40 GB HBM, with dropout and a padding mask. We compare against reference implementations for exact attention, approximate attention, and sparse attention. We report a subset of baselines in the main body; Appendix E contains more baselines and full details.

[4]Path-256 requires longer sequences but has relatively shorter paths than Path-X, so it is easier to obtain a higher accuracy.